Chapter 6 Introduction to Multiple Regression

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Statistical Decision Making (45-752)

[24]:
```
import tprstats
import pandas
import matplotlib.pyplot as plt
import numpy
from scipy.stats import t

# Jupyter "magic" command
# precision - round floats to 4 significant digits
%precision %.4f
```

[24]: '%.4f'

1 Introduction to Multiple Regression

1.0.1 Learning objectives

After completing this chapter, you will be able to:

- Explain how a multiple regression model represents a relationship among variables.
- Estimate a multiple regression model.
- Interpret the results of a multiple regression model.
- Make predictions using a multiple regression model.
- Plot visual diagnostics to assess the goodness of fit of a model to the data.

1.1 The Multiple Regression Model

The multiple regression model is a relationship between a dependent variable y and K explanatory variables x_1, x_2,...,x_K.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_K x_K + \varepsilon$$

In the above equation, the coefficients β_0, β_1,...,β_K are unknown constants that we would like to estimate. Recall that, in statistics, we refer to such unknown constants as parameters. The error term is ε.

Suppose that we have a sample of n observations of data from the model. We will index observations by i: $i = 1, ..., n$. The equation for an observation i is:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + ... + \beta_K x_{Ki} + \varepsilon_i$$

We will index variables by k: $k = 1, ..., K$. Both n and K will depend on your application. Also, it is useful for you to know that the right-hand-side variables are sometimes referred to as explanatory variables and sometimes as independent variables. The above equation is our population model.

In our discussion of the simple regression model, we introduced notation for the estimated coefficients, predicted values, and estimated error terms. The following extends this notation to the multiple regression model. The estimated coefficient estimates are denoted b_0, b_1,...,b_K.

Using these estimated coefficients, we obtain the fitted value for each observation:

$$\hat{y}_i = b_0 + b_1 x_{1i} + b_2 x_{1i} + ... + b_K x_{Ki}$$

We then calculate the estimated error term, e_i, for each observation:

$$e_i = y_i - \hat{y}_i$$

Brief recap:

- β_k denotes a parameter (i.e., an unknown population coefficient).
- b_k denotes an estimated coefficient.
- ε_i denotes the error term for observation i in the population model.
- e_i denotes the estimated error term for observation i.

1.2 Investigate an Example

We begin with a simple example using data that I made up. I put the data in an Excel file named Reg_Example.xlsx. In this example, there are two explanatory variables, x_1 and x_2. Hence, $K = 2$ in this example. There are 20 observations in the sample. Thus, $n = 20$ in this application.

Import the file Reg_Example.xlsx.

1.3 Let's Study the Data

[25]:
```
Reg_Example = pandas.read_excel("data/Reg_Example.xlsx")
```

View the data. When viewing a dataframe, an unlabeled column containing the *indices* is also included.[1] Notice that Observation 1 is at index $i = 0$. This is because Python counts from 0 instead of 1.

[26]:
```
Reg_Example
```

[26]:

	Observation	y	x1	x2
0	1	24.42421	2.572799	16.624739
1	2	17.99297	2.247157	9.665573

[1] Indices are labels that help us organize, access and manipulate data efficiently.

```
2             3  16.42750  6.349159   6.267691
3             4  24.44181  7.966056   6.213770
4             5  24.02260  4.546715   8.721013
5             6  37.35047  7.883281  13.103613
6             7  20.87542  6.483221   5.431699
7             8  27.55375  5.468359  14.156530
8             9  25.56716  6.991105   8.560674
9            10  26.55874  6.129744  16.384932
10           11  12.53338  3.183348   6.074271
11           12  28.42547  7.024169   9.975649
12           13  23.71417  3.873335  12.213256
13           14  30.93722  8.580459  11.696897
14           15  14.94540  2.576798   6.618587
15           16  24.36723  7.920643  11.794463
16           17  31.83901  8.333082   8.684492
17           18  29.05402  5.261493  17.356139
18           19  23.32666  4.155327  15.811660
19           20  33.25426  7.358152  17.664795
```

If we want to access a row by index, we use the iloc() command. "iloc" is short for integer location. If we lookup observation 12 located at index $i = 11$, we get the values x1=7.024169, x2=9.975649, and y=28.42547.

[27]:
```
Reg_Example.iloc[11]
```

[27]:
```
Observation    12.000000
y              28.425470
x1              7.024169
x2              9.975649
Name: 11, dtype: float64
```

Let's also view summary statistics of the data. For convenience in the discussion in the following text, I round to two digits after the decimal point. We see from the summary statistics that the range of x1 is [2.25,8.58], the range of x2 is [5.43,17.66,], and the range of y is [12.53,37.35]. Knowing the ranges of these variables will be useful for our discussion below. Keep in mind that the smallest values of x1 and x2 are 2.25 and 5.43 respectively.

[28]:
```
Reg_Example.describe()
```

[28]:
```
       Observation          y         x1         x2
count     20.00000  20.000000  20.000000  20.000000
mean      10.50000  24.880572   5.745220  11.151022
std        5.91608   6.250930   2.083769   4.130966
min        1.00000  12.533380   2.247157   5.431699
25%        5.75000  22.713850   4.084829   8.075152
50%       10.50000  24.433010   6.239451  10.836273
75%       15.25000  28.582608   7.489434  14.570312
max       20.00000  37.350470   8.580459  17.664795
```

For practice view histograms of y, x1, and x2. You will see that there are no prominent outliers.

1.3.1 Estimate a Regression Using Data from Our Example

Next, let's estimate a regression of y against x1 and x2 and name the result RegModel. I named the result RegModel. Note that the independent variables, x1 and x2, are separated by a plus sign. This is how we tell Python that we want both x1 and x2 to be included as variables in the regression.

[29]:
```
RegModel = tprstats.model("cs", "y~x1+x2", Reg_Example)
```

1.3.2 Interpret the Results and Investigate Statistical Significance

View a summary of the regression output using the summary command. We see that the p-value for x1 is less than .05. Hence, we conclude that x1 is statistically significant at the 5% level. Please answer the following four questions. Then view the footnote to check your answers.

- Is x2 statistically significant at the 5% level?
- Is x2 statistically significant at the 1% level?
- Suppose x1 increases by one unit and x2 is unchanged. Does the predicted value of y increase by 2.024?
- Suppose x2 increases by one unit and x1 is unchanged. What is the predicted change in y?[2]

[30]:
```
RegModel.summary()
```

[30]:

Dep. Variable:	y	**R-squared:**	0.773
Model:	OLS	**Adj. R-squared:**	0.747
No. Observations:	20	**F-statistic:**	32.90
Covariance Type:	HC1	**Prob (F-statistic):**	1.43e-06

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	3.4004	2.522	1.348	0.195	-1.921	8.722
x1	2.0240	0.321	6.312	0.000	1.347	2.701
x2	0.8835	0.144	6.154	0.000	0.581	1.186

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)

1.4 Visualualize A Regression with Two Explanatory Variables

A regression with two explanatory variables from a different data set is displayed on a three-dimensional plot below. When I first began to study regression, I thought a regression with two explanatory variables would be a line in three dimensions. I was wrong. The regression equation is a plane in three dimensions. The dependent variable, shown on the vertical axis is named y. The explanatory variables are named x1 and x2. The coefficients of the two explanatory variables are b1 and b2 respectively. Study the labeling of the axes carefully. Be sure to see that the coefficients of the regression are the slopes where the regression plane intersects the two "walls" of the graph.

[2] The p-value for x2 is less than .01. Hence, x2 is significant at both the 5% level and the 1% level. If x1 increases by one unit and x2 is unchanged, the predicted value of y increase by 2.024. If x2 increases by one unit and x1 is unchanged, y is predicted to increase by .88.

In this model, we have two explanatory variables. Hence the population equation for an observation i is:

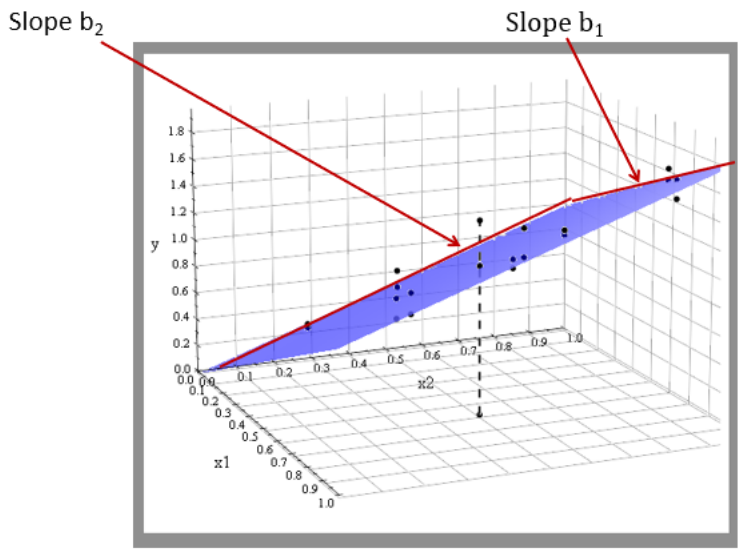
$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i$$

Our estimated model is:

$$\hat{y}_i = b_0 + b_1 x_{1i} + b_2 x_{1i}$$

1.4.1 Visualize Slope Coefficients

Do you agree that both estimated coefficients equal one, i.e., $b_1 = 1$ and $b_2 = 1$? When you have decided on your answer, check the footnote.[3]

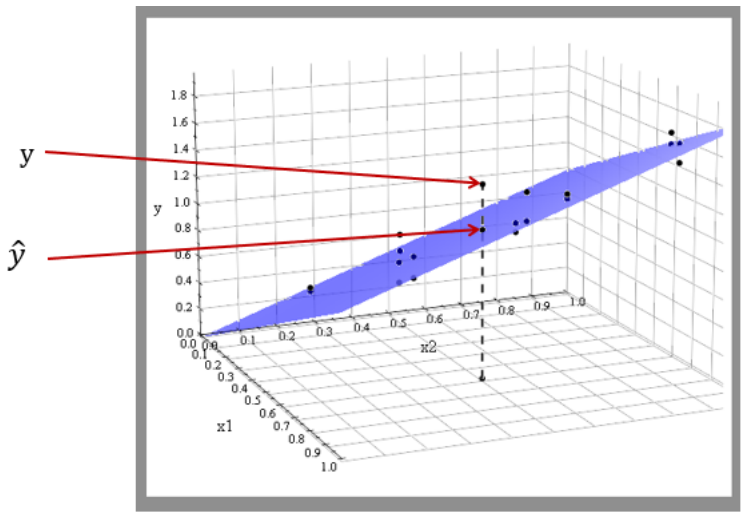


1.4.2 Visualize Actual and Fitted Values

The plot below shows the same data and regression equation as the previous image. In this graph, I have added a vertical line to illustrate the actual value and the fitted value for one observation. The fitted value is the point where the vertical line intersects the regression plane. What are the values of x1 and x2 for this observations. When you have decided on your answer, check the footnote.[4]

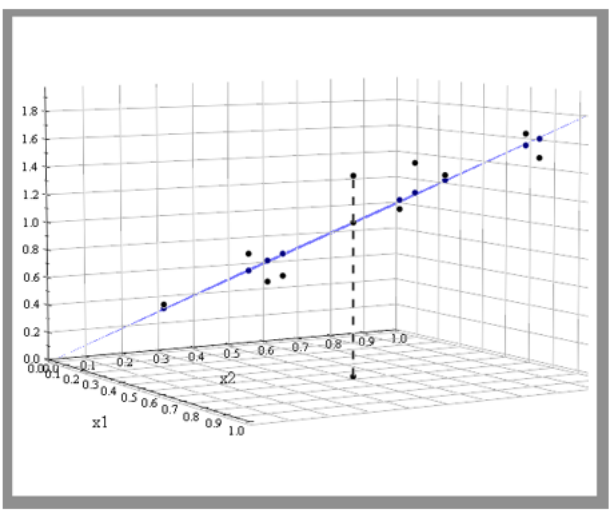
[3] Yes

[4] Answer: x1=.6 and x2=.5.



1.5 Visualize Plane Rotated to Show All Estimated Error Terms

I rotated the plot so that we can easily see the actual and fitted values of for all the observations. For each observation (x_{1i}, x_{1i}), the estimated error term is the vertical distance between the observed value of the dependent variable, y_i, and the predicted value $\hat{y}_i$. The vertical dashed line is through the same point depicted on the previous graph. When the observed value, y_i, is above the regression plane, the estimated error term is positive. When the observed value is below the regression plane, the estimated error term is negative.



1.6 Indicator and Continuous Variables: Parcel Delivery

Multiple regression models have broadly two types of variables, continuous variables and indicator variables. Above we estimated a model with two continuous explanatory variables. An indicator variable can take on values of either 0 or 1. We will now study a model with a continuous variable and indicator variables.

Import the Parcel_Delivery.xlsx dataset.

[31]:
```
Parcel_Delivery = pandas.read_excel("data/Parcel_Delivery.xlsx")
```

This data set has information about delivery time on three different delivery routes for a parcel delivery company named OTD (On Time Delivery). In this dataset, the three different delivery routes are denoted RA, RB, and RC. If the delivery was on Route A, then RA=1, otherwise RA=0. RB and RC are defined similarly. For each of those days, we know the number of parcels delivered (PARCELS) and the time that it took the driver to deliver the parcels (DELTIM). This includes the time that it took the driver to drive the delivery van from the depot to the route, deliver the parcels, and return the delivery van to the depot. For the time being, ignore the columns in the data set headed ROUTE_NAME and ROUTE_NUMBER.

A subsample of the data is shown below. You will see that the first two rows in the file are deliveries on Route A and the second two rows are deliveries on Route B. The first row tells us that 53 parcels were delivered that day with a delivery time of 414 minutes–almost 7 hours. On which routes were the deliveries in each of rows 5, 6, and 7? When you have decided on your answer, view the solution in the footnote.[5]

	DELTIM	PARCELS	RA	RB	RC
1	414	53	1	0	0
2	330	44	1	0	0
3	349	45	0	0	1
4	326	41	0	0	1
5	176	18	1	0	0
6	154	13	0	0	1
7	111	13	0	1	0

1.6.1 Parcel Delivery Regression and Interpretation

[32]:
```
ParcelReg = tprstats.model("cs", "DELTIM~PARCELS+RB+RC", Parcel_Delivery)
ParcelReg.summary()
```

[32]:

Dep. Variable:	DELTIM	**R-squared:**	0.976
Model:	OLS	**Adj. R-squared:**	0.975
No. Observations:	100	**F-statistic:**	1488.
Covariance Type:	HC1	**Prob (F-statistic):**	2.58e-80

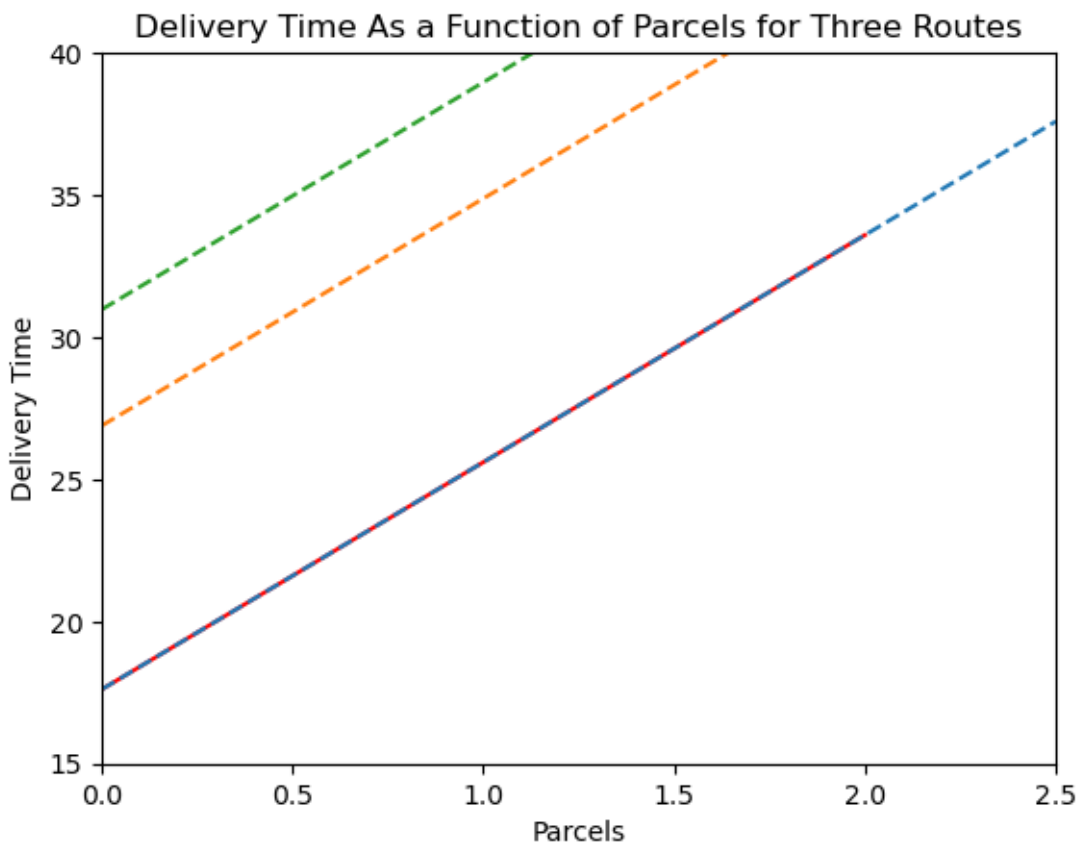
	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	17.6083	5.767	3.053	0.003	6.160	29.057
PARCELS	7.9919	0.120	66.479	0.000	7.753	8.231
RB	9.2761	4.973	1.865	0.065	-0.595	19.148
RC	13.3647	4.877	2.740	0.007	3.683	23.046

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)

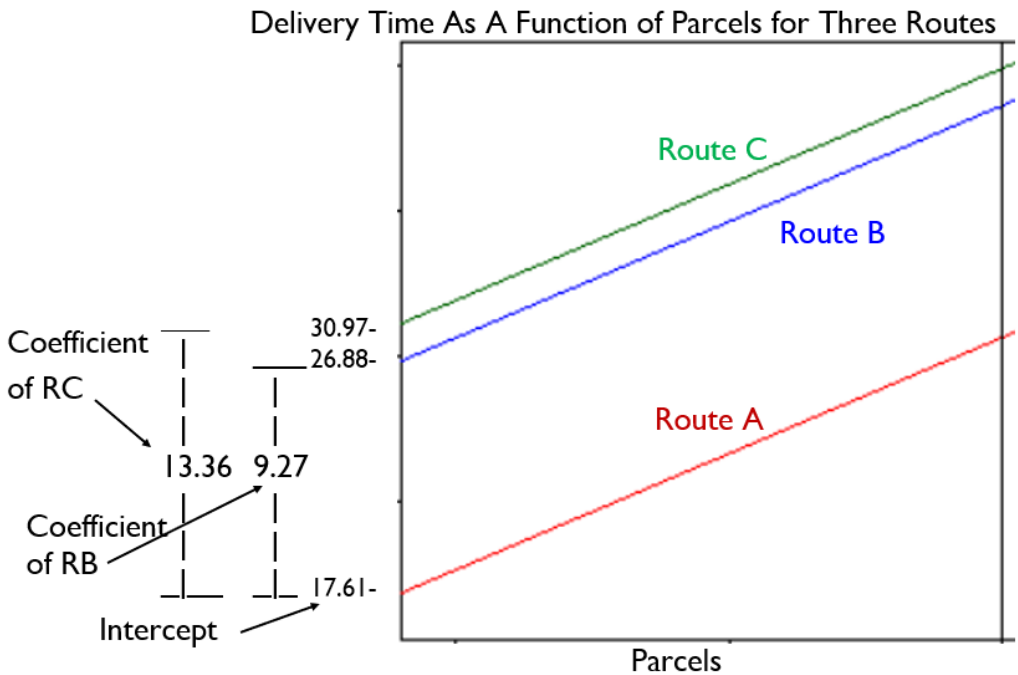
[5] Delivery 5 was on Route A. Delivery 6 was on Route C. Delivery 7 was on Route B.

[33]:
```
Xnew = pandas.DataFrame({"PARCELS": [0, 2], "RB": [0, 0], "RC": [0, 0]})

y_pred = ParcelReg.predict(Xnew)
plt.plot(Xnew["PARCELS"], y_pred, color="red")
plt.xlim(0, 2.5)
plt.ylim(15, 40)


def abline(intercept, slope):
    axes = plt.gca()
    x_vals = numpy.array(axes.get_xlim())
    y_vals = intercept + slope * x_vals
    plt.plot(x_vals, y_vals, "--")


abline(ParcelReg.params.iloc[0], ParcelReg.params.iloc[1])
abline((ParcelReg.params.iloc[0] + ParcelReg.params.iloc[2]), ParcelReg.params.
 ↪iloc[1])
abline((ParcelReg.params.iloc[0] + ParcelReg.params.iloc[3]), ParcelReg.params.
 ↪iloc[1])

plt.xlabel("Parcels")
plt.ylabel("Delivery Time")
plt.title("Delivery Time As a Function of Parcels for Three Routes")
```

[33]:
```
Text(0.5, 1.0, 'Delivery Time As a Function of Parcels for Three Routes')
```



The graph below displays the model estimated above. Consider first a delivery on Route A. For a delivery on Route A, variables RB and RC are zero. Thus, the coefficients of the last two terms are multiplied by zero. Hence, for a delivery on Route A, the two terms that remain are the intercept and the coefficient of parcels. The intercept is the fixed cost in minutes for the driver to drive from the depot to Route A at the beginning of the day plus the time to drive from Route A to the depot at the end of the day. From the regression above, you can see that the fixed cost is estimated to be 17.61 minutes for Route A. The coefficient of PARCELS tells us, when the driver is on the route, that the estimated time for each additional parcel is 7.99 minutes.



Next, consider a delivery on Route B. For a delivery on Route B, RB=1 and RC=0. Hence, the first three coefficients of our regression apply for Route B. The fixed time for Route B is 9.27 minutes more than for Route A. Note where this difference is displayed on the graph above. Hence, the total fixed time on Route B is (17.61+9.27)=26.88 minutes. When the driver is on the route, the coefficient of PARCELS tells us that the estimated time for each additional parcel is 7.99 minutes.

Next, consider a delivery on Route C. For a delivery on Route C, RC=1 and RB=0. Hence, the first, second, and fourth coefficients apply for Route C. The fixed time for Route C is 13.36 minutes more than for Route A. Note where this difference is displayed on the graph above. Hence, the total fixed time on Route C is (17.61+13.36)=30.97 minutes. When the driver is on the route, the coefficient of PARCELS again tells us that the estimated time for each additional parcel is 7.99 minutes.

In the above example, the reference route is Route A. The coefficient for RB is the estimated difference in fixed time between routes B and A. The coefficient for RC is the estimated difference in fixed time between routes C and A. Be sure to study this example carefully. I think you will find the graphical display very helpful in understanding models with indicator and continuous variables. In this example, we have assumed that the slope coefficient (marginal cost in minutes for each additional package) is the same for all three routes. Later, we will see how to allow the slope coefficients to differ across routes.

1.6.2 Parcel Delivery: Regression Using The C() Command

When there are only three delivery routes, it is easy to create separate indicator variables for the routes. In practice, delivery services have a large number of routes in an urban area, and the designation of delivery routes in a data set would typically be a single column. That column might designate routes by name, as in the ROUTE_NAME column in our data set. View the dataset so that you seen this variable in the file. We can use this variable along with the `C()` option to create indicator variables. The parcel delivery regression with factors based on ROUTE_NAME is shown

below. Note the use of `C(ROUTE_NAME)` in the regression. Then compare the regression results to those obtained above. You will see that the results are identical.

This illustrates the convenience of the `C()` option. One downside to the `C()` option is how it labels indicator variables. The indicator variable for route B is labeled `C(ROUTE_NAME)[T.RB]` instead of just `RB`.

[34]:
```
ParcelRegFac = tprstats.model("cs", "DELTIM~PARCELS+C(ROUTE_NAME)",␣
↪Parcel_Delivery)
ParcelRegFac.summary()
```

[34]:

Dep. Variable:	DELTIM	**R-squared:**	0.976
Model:	OLS	**Adj. R-squared:**	0.975
No. Observations:	100	**F-statistic:**	1488.
Covariance Type:	HC1	**Prob (F-statistic):**	2.58e-80

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	17.6083	5.767	3.053	0.003	6.160	29.057
PARCELS	7.9919	0.120	66.479	0.000	7.753	8.231
C(ROUTE_NAME)[T.RB]	9.2761	4.973	1.865	0.065	-0.595	19.148
C(ROUTE_NAME)[T.RC]	13.3647	4.877	2.740	0.007	3.683	23.046

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)

In practice, routes might be denoted by number instead of name. To illustrate this alternative, variable ROUTE_NUMBER denotes the routes by number. The `C()` option also works when alternatives are designated by number as below. Note that the estimates are the same as in the above regression.

[35]:
```
ParcelRegFac = tprstats.model("cs", "DELTIM~PARCELS+C(ROUTE_NUMBER)",␣
↪Parcel_Delivery)
ParcelRegFac.summary()
```

[35]:

Dep. Variable:	DELTIM	**R-squared:**	0.976
Model:	OLS	**Adj. R-squared:**	0.975
No. Observations:	100	**F-statistic:**	1488.
Covariance Type:	HC1	**Prob (F-statistic):**	2.58e-80

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	17.6083	5.767	3.053	0.003	6.160	29.057
PARCELS	7.9919	0.120	66.479	0.000	7.753	8.231
C(ROUTE_NUMBER)[T.2]	9.2761	4.973	1.865	0.065	-0.595	19.148
C(ROUTE_NUMBER)[T.3]	13.3647	4.877	2.740	0.007	3.683	23.046

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)

As noted above, a delivery service will typically operate a large number of routes in a metropolitan area. The `C()` option does the work for us regardless of the number of delivery routes.

For practice, estimate the parcel delivery model with Route C as the reference category using the following regression command. Then interpret the coefficients of RA and RB.

```
ParcelReg2=tprstats.model("cs", "DELTIM~PARCELS+RA+RB", Parcel_Delivery)
```

1.6.3 Managerial Application

What is the managerial value of a model like the parcel delivery model? Such a model can be used to help lay out routes so that the total delivery time on each is roughly the same. The objective would be to lay out routes so that no driver is burdened by excessively long delivery days while also avoiding having routes where the time of drivers might be underutilized. This type of model can also be used in other settings. For example, some time ago, a Tepper student worked with the sanitation department of a city to help balance routes for trash collection. Again, the objective was to avoid having routes that required excessively long days for some or inefficiently short days for others. More generally, statistical models coupled with operations research optimization methods can be used for efficient design of delivery routes.

1.7 Analyze Purchase Price of Tires for Leading Auto Manufacturer

Equipping a vehicle with tires is a significant expense item for automobile manufacturers. A Tepper alum working for an auto manufacturer developed a statistical model for his company to predict prices of tires. The motivations for developing the model were the following:

- Predict price to aid purchasing team in negotiation with suppliers
- Predict price for automobile designers of the cost of equipping a car with a particular type of tire

The company approved providing the data and statistical model to us. Our alum collected data on recent tire purchases by the company. A description of the variables in the data file follows:

Variable to Be Predicted: Price (Dollars per Tire)

Tire Characteristics:

- Diameter: Inside diameter of the tire in inches
- Ratio: Ratio of height of tire sidewall to width of tread, in percent.
- Terrain: Equals 1 if all tire is for all Terrain vehicle, 0 otherwise
- Weight: Weight of tire in pounds
- Tread: Tread width in millimeters
- Volume: The number of tires of this type purchased by the company in the reference year (thousands)

The speed safety classification of each tire. A tire has one of four possible speed ratings:

- R: rated for normal highway driving
- H: rated to 90 miles per hour
- V: rated to 120 miles per hour
- Z: rated to 150 miles per hour

Type of Exterior. A tire has one of three possible exteriors:

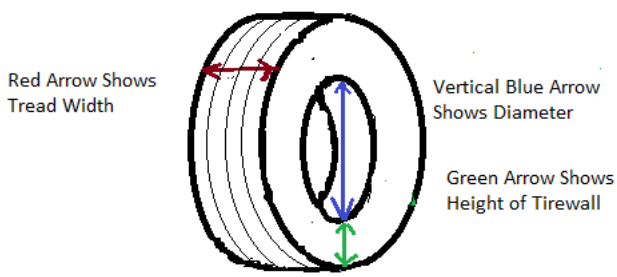
- BW: Black Wall
- WW: White Wall

- OWL: Outside White Lettering

Please import Tire_Pricing.xlsx.

[36]:
```
Tire_Pricing = pandas.read_excel("data/Tire_Pricing.xlsx")
```

The sketch below shows diameter, tread, and height of sidewall. The variable Ratio is the height of the sidewall divided by the width of the tread expressed in percent.



1.7.1 Study the Tire Data Set

The first twelve observations in the file are shown below. Note that variables "Rating" and "Tirewall" contain names rather than numbers. Note also, that indicator variables variables are included that denote speed rating and type of exterior. Thus, the information is entered in two different ways.

View observation 9. You will see that this was a purchase of a blackwall tire with speed rating for normal highway driving. You will also see that the company purchased 2,028,000 of these tires at a price of $68.79 per tire. Thus, the total expenditure for purchase of the tires in observation 9 was $139.5 million dollars.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Observation	Price	Diameter	Ratio	Tread	Weight	Rating	Tirewall	Terrain	Volume	R	H	V	Z	BW	WW	OWL
2	1	91.17	16	60	225	24.6	R	BW	0	143	1	0	0	0	1	0	0
3	2	77.7	15	60	205	21.6	R	BW	0	473	1	0	0	0	1	0	0
4	3	105.39	16	60	215	24.2	V	BW	0	49	0	0	1	0	1	0	0
5	4	128.79	17	50	235	26.5	V	BW	0	120	0	0	1	0	1	0	0
6	5	128.82	16	60	215	24	Z	WW	0	27	0	0	0	1	0	1	0
7	6	91.17	15	65	205	21.5	H	BW	0	21	0	1	0	0	1	0	0
8	7	59.82	14	70	185	17	R	BW	0	208	1	0	0	0	1	0	0
9	8	99.39	16	60	215	23.8	H	BW	0	62	0	1	0	0	1	0	0
10	9	68.79	15	65	205	20	R	BW	0	2028	1	0	0	0	1	0	0
11	10	84.33	15	65	205	20	H	BW	0	10	0	1	0	0	1	0	0
12	11	75.93	15	70	215	20.8	R	BW	0	396	1	0	0	0	1	0	0

View summary statistics for numeric variables by using the describe command. We see that Price per tire ranged from $60 to $152 and Diameter ranged from 14 inches to 17 inches. What was median diameter? What was the largest number of tires acquired in a single purchase?

[37]:
```
Tire_Pricing.describe().transpose()
```

[37]:

	count	mean	std	min	25%	50%	75%	\
Obs	61.0	31.000000	17.752934	1.00	16.00	31.00	46.00	
Price	61.0	99.687541	23.533104	59.82	78.27	100.35	117.66	
Diameter	61.0	15.540984	0.807749	14.00	15.00	16.00	16.00	
Ratio	61.0	64.836066	7.301097	45.00	60.00	65.00	70.00	
Tread	61.0	222.868852	20.503232	185.00	215.00	225.00	235.00	
Weight	61.0	25.180328	5.609570	16.20	21.50	24.30	27.70	
Terrain	61.0	0.131148	0.340363	0.00	0.00	0.00	0.00	
Volume	61.0	254.229508	387.708584	5.00	49.00	120.00	288.00	
R	61.0	0.704918	0.459865	0.00	0.00	1.00	1.00	
H	61.0	0.114754	0.321370	0.00	0.00	0.00	0.00	
V	61.0	0.098361	0.300273	0.00	0.00	0.00	0.00	
Z	61.0	0.081967	0.276591	0.00	0.00	0.00	0.00	
BW	61.0	0.950820	0.218039	0.00	1.00	1.00	1.00	
WW	61.0	0.049180	0.218039	0.00	0.00	0.00	0.00	
OWL	61.0	0.196721	0.400819	0.00	0.00	0.00	0.00	

	max
Obs	61.00
Price	151.56
Diameter	17.00
Ratio	75.00
Tread	265.00
Weight	40.80
Terrain	1.00
Volume	2028.00
R	1.00
H	1.00
V	1.00
Z	1.00
BW	1.00
WW	1.00
OWL	1.00

Calculate the total number of tires purchased by the company in the year that our alum studied. Volume is expressed in thousands. The company purchased 15.508 million tires in the year our alum studied.

[38]:
```
Tire_Pricing["Volume"].sum()
```

[38]:
```
np.int64(15508)
```

Calculate total expenditure made by the company for tires for the year our alum studied. The company$1.41 billion on tires in the year our alum studied. Tires are a major purchase item for the auto manufacturer!

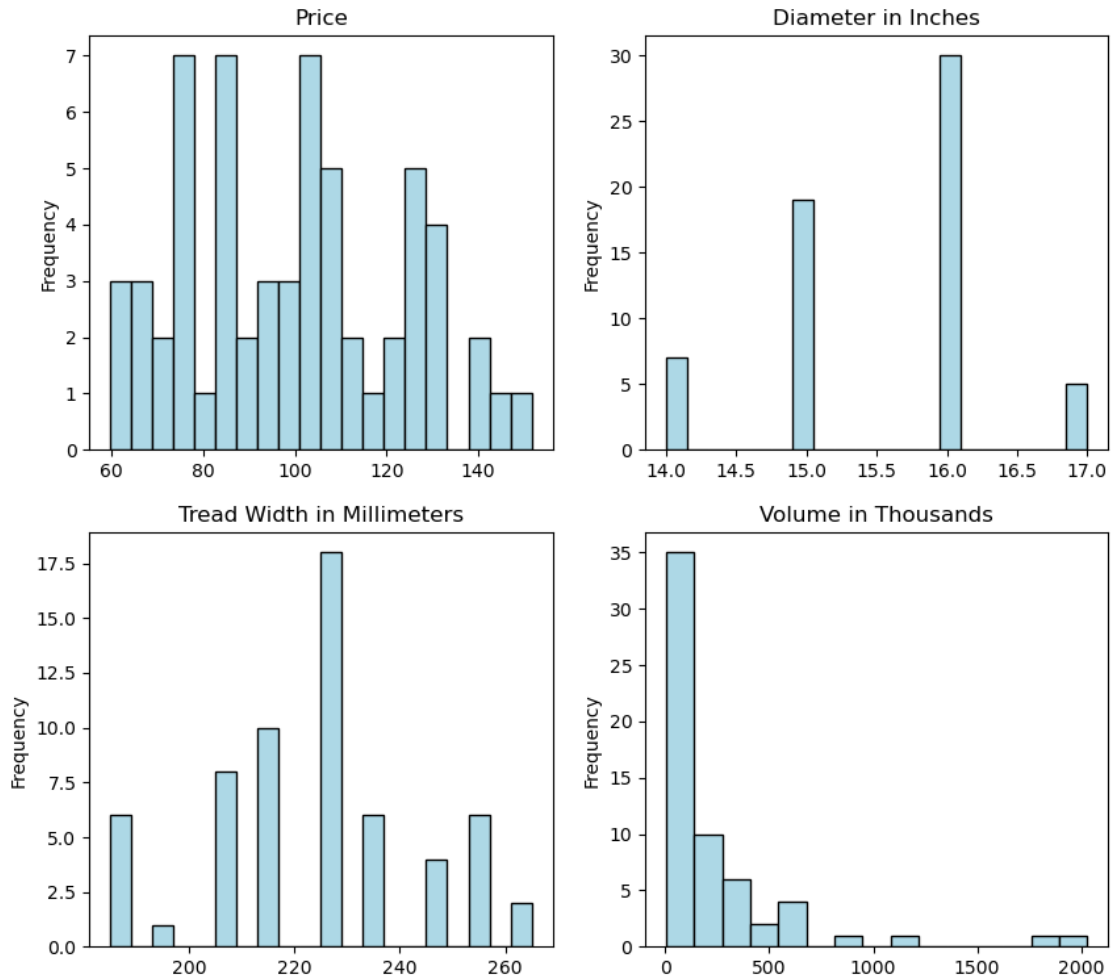
[39]:
```
total_expenditures = Tire_Pricing[["Price", "Volume"]].product(axis=1).sum()
total_expenditures
```

[39]: 1409865.6300

1.7.2 View Histograms of Variables

We see that the distributions of Prices, Diameter, and Tread are relatively symmetric while the distribution of Volume is skewed to the right. Large volume purchases were made less frequently than small volume purchases.

[40]:
```
fig, axes = plt.subplots(nrows=2, ncols=2)
Tire_Pricing["Price"].plot.hist(
    ax=axes[0, 0],
    bins=20,
    title="Price",
    edgecolor="black",
    color="lightblue",
    figsize=(10, 9),
)
Tire_Pricing["Diameter"].plot.hist(
    ax=axes[0, 1],
    bins=20,
    title="Diameter in Inches",
    edgecolor="black",
    color="lightblue",
    figsize=(10, 9),
)
Tire_Pricing["Tread"].plot.hist(
    ax=axes[1, 0],
    bins=20,
    title="Tread Width in Millimeters",
    edgecolor="black",
    color="lightblue",
    figsize=(10, 9),
)
Tire_Pricing["Volume"].plot.hist(
    ax=axes[1, 1],
    bins=15,
    title="Volume in Thousands",
    edgecolor="black",
    color="lightblue",
    figsize=(10, 9),
)
plt.show()
```



You might also wish to do histograms of Weight and Ratio.

1.8 Tire Pricing Regression Model

The regression estimated by our alum:

[41]:
```
TireReg = tprstats.model(
    "cs", "Price~Diameter+Ratio+Terrain+Weight+Tread+Volume+WW+OWL+H+V+Z",␣
 ↪Tire_Pricing
)
TireReg.summary()
```

[41]:

Dep. Variable:	Price	**R-squared:**	0.933
Model:	OLS	**Adj. R-squared:**	0.918
No. Observations:	61	**F-statistic:**	106.2
Covariance Type:	HC1	**Prob (F-statistic):**	2.99e-30

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	-27.2056	52.689	-0.516	0.608	-133.089	78.678
Diameter	5.3728	2.780	1.933	0.059	-0.213	10.959
Ratio	-0.6184	0.271	-2.284	0.027	-1.163	-0.074
Terrain	8.9199	3.203	2.785	0.008	2.483	15.357
Weight	1.8909	0.627	3.018	0.004	0.632	3.150
Tread	0.1305	0.167	0.782	0.438	-0.205	0.466
Volume	-0.0033	0.002	-1.775	0.082	-0.007	0.000
WW	4.4036	2.738	1.608	0.114	-1.099	9.907
OWL	4.0631	3.127	1.299	0.200	-2.222	10.348
H	8.0769	2.412	3.348	0.002	3.229	12.924
V	23.8924	4.204	5.684	0.000	15.445	32.340
Z	26.3225	5.772	4.561	0.000	14.724	37.921

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)
[2] The condition number is large, 2.96e+04. This might indicate that there are strong multicollinearity or other numerical problems.

This model has multiple categories of indicator variables: Terrain has two categories, yes if all-terrain, no if not all-terrain. Exterior color has two categories, BW or WW. Outside white lettering has two categories, yes or no. Speed rating has four categories (R,H,V,Z) The reference chosen for each category is a matter of convenience and can easily be inferred: In the above regression, the reference category for all-terrain is a tire that is not all-terrain. The reference for OWL is no OWL. Black wall is the most common type of exterior. It is natural to choose BW as the reference category for exterior color as our alum did. The most common speed rating is a tire rated for normal highway driving, R. Hence, it is natural to make type R tire the reference among the four categories: (R,H,V,Z). The model also has multiple continuous variables: Diameter, Ratio, Weight, Tread, Volume.

For practice, estimate the model with BW and without WW. Then compare the coefficients to the model above. You will see that the coefficient of BW will be equal in magnitude to the above coefficient for WW and opposite in sign. Also, BW will have the same p-value. Explain the intuition. You will also see that the intercept differs between the two regressions. Explain the intuition.

1.8.1 Interpret The Estimated Model

Recall three Key Criteria for evaluating regression results:

- Significance: Are the variables statistically significant?
- Sign: Do the algebraic signs make sense for variables that are significant?
- Magnitude: Are the magnitudes of the significant coefficients plausible?

WW, OWL, Tread are not very significant.

Three possible reasons:

- Those variables don't affect tire prices.
- Those variables do affect tire prices but we don't have enough observations or variation to get precise estimates. (Verify that 3 tires have WW, 12 have OWL, none have both.)
- The variables are highly correlated with each other making it difficult to distinguish their separate effects.

Should you keep or drop WW, OWL, or Tread?

- If you were in the industry, you would be able to judge whether there is an extra cost to OWL or WW and whether $4 is a reasonable estimate of the cost. (It seems reasonable to me.)
- Tread and weight are relatively highly correlated (.91). As an industry expert, you could judge whether to keep both. If the coefficients are reasonable in magnitude, there is no harm in retaining them (i.e., dropping one or the other will not make your prediction more precise).
- Our alum judged that it was best to retain these variables, and I agree.

Use the elasticities command to find the elasticities of price with respect to the explanatory variables. You will find the elasticity of price with respect to diameter is .838. Interpret the result. Then look at the footnote to check your answer[6] The elasticities with respect to the indicator variables are not very meaningful. For example, it is not possible to increase Terrain by one percent.

All of the variables in the regression are characteristics of tires except the Volume variable. Is the volume discount large enough to be managerially important? Let's investigate.

- The coefficient of volume in the regression is -.0033.
- Average volume per purchase is 250 thousand tires. How do we know this? When you have decided on your answer, check the footnote.[7]
- Hence, volume discount **per tire** on an average purchase is $250 * 0.0033 = 0.825$. This is $0.825.
- Thus, on a purchase of 250,000 tires, the volume discount savings would be the number of tires purchased time savings per tire: $250,000 * 0.825 = 206,000$

This is indeed large enough to be managerially important!

1.8.2 Investigate the residuals from the model

It is generally good practice to investigate the residuals in a regression. One reason to do this is to investigate whether there are outliers. We will also see going forward that investigating residuals is useful in assessing functional form.

Let's Calculate and plot residuals. We extract the residuals using the first command below. I named the result Resids. You can choose any name you like. The second command below specifies a rectangular plot. The third command plots the residuals against the observation number from the file, which is named Obs. The fourth command connects the dots with lines. The fifth adds a horizontal line at zero.

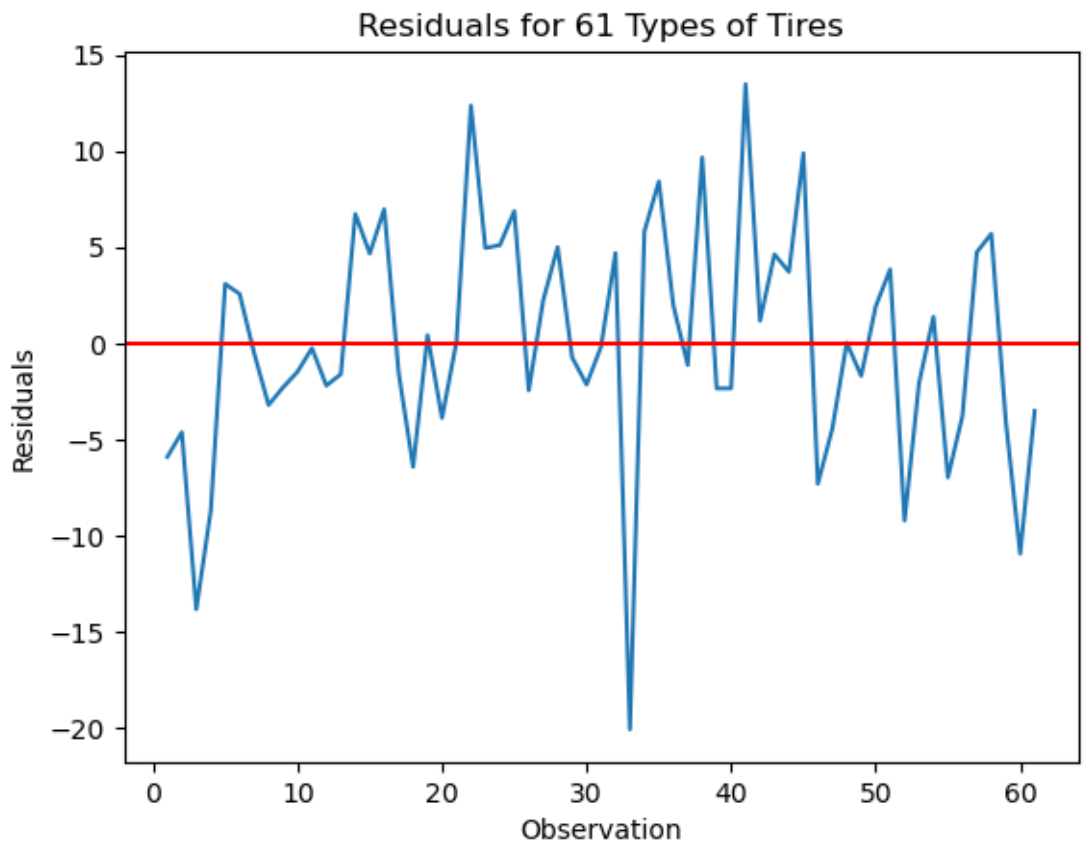
[42]:
```
resids = TireReg.resid
plt.plot(Tire_Pricing["Obs"], resids)
```

[6] Holding all other explanatory variables constant, a one percent increase in diameter increases predicted price by .838 percent.

[7] We know this from the summary statistics.

```
plt.axhline(0, color="red")
plt.title("Residuals for 61 Types of Tires")
plt.xlabel("Observation")
plt.ylabel("Residuals")
```

[42]: Text(0, 0.5, 'Residuals')



Find the index of the large negative residual. We find that the large negative residual is for tire 33. Look at the data file and you will see this is a Z-rated tire.

[43]:
```
index = resids.argmin()
Tire_Pricing.iloc[index]
```

[43]:
```
Obs             33
Price       106.53
Diameter        16
Ratio           50
Tread          215
Weight        23.5
Rating           Z
```

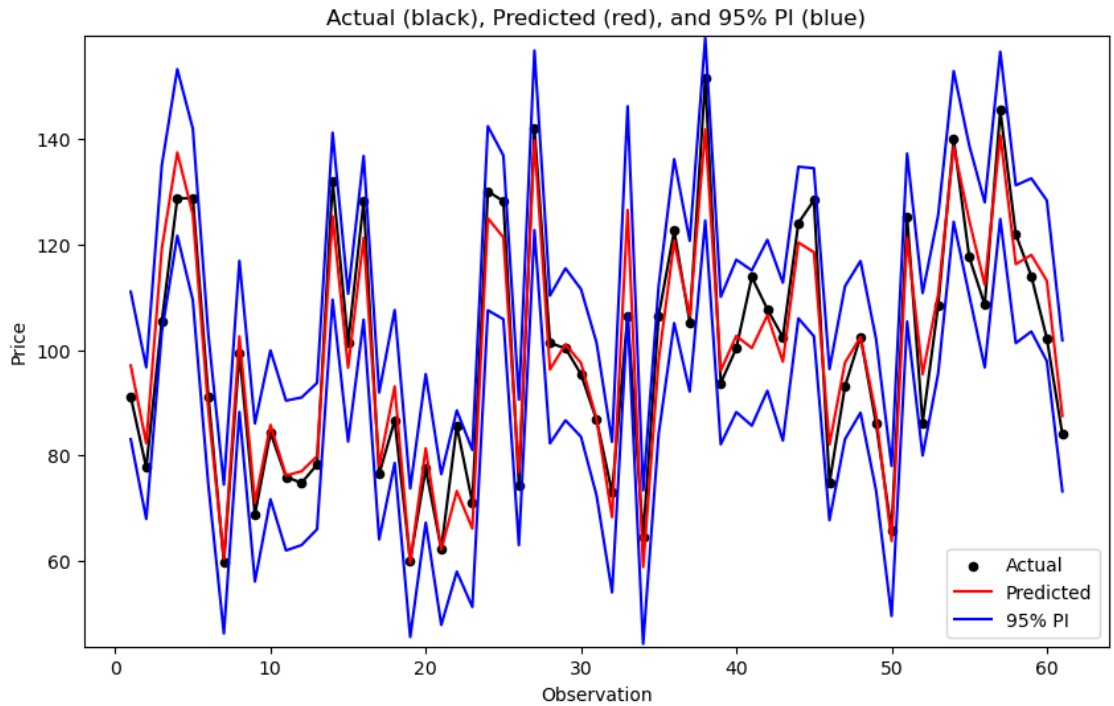
```
Tirewall        BW
Terrain          0
Volume          20
R                0
H                0
V                0
Z                1
BW               1
WW               0
OWL              0
Name: 32, dtype: object
```

Select all Z tires and view to see if there is anything unusual about this tire. The tire in question is the fourth row below. We see nothing unusual about this tire. It appears the company struck a good bargain for these 20 tires.

Obs	Price	Diameter	Ratio	Tread	Weight	Rating	Tirewall	Terrain	Volume	R	H	V	Z	BW	WW	OWL
5	128.82	16	60	215	24.0	Z	WW	0	27	0	0	0	1	0	1	0
24	130.11	16	55	225	23.6	Z	BW	0	16	0	0	0	1	1	0	0
27	142.02	17	45	245	24.1	Z	BW	0	120	0	0	0	1	1	0	0
33	106.53	16	50	215	23.5	Z	BW	0	20	0	0	0	1	1	0	0
38	151.56	17	45	245	25.1	Z	BW	0	50	0	0	0	1	1	0	0

The following is another plot that can sometimes be informative. For each observation in the file, it plots the actual and fitted values of the dependent variable along with a 95% prediction interval for each. If you look very closely, you may be able to see that the actual value for observation 33 is very near the lower boundary of the 95% prediction interval, well below the fitted value shown in red. The plot above with just the residuals proves to be more helpful in highlighting this outlier.

[44]:
```
TireReg.plot_actual_fitted()
```



1.8.3 Predict price of a tire for design team.

Application: The design team in the company is anticipating using a tire with the following characteristics for a car they are designing.

The tire would have a diameter of 15, ratio of 65, weight of 25, and tread width of 225. The tire would be speed-rated for normal highway driving (R), would not be all terrain, and would not have white walls or outside white lettering. If this tire configuration is adopted for the car, the company expects to purchase approximately 300,000 of the tires each year. The design team would like to have an estimate of the price that the company can expect to pay for the tire, and they would like a price range that they can be 75% sure of achieving when negotiating the purchase.

[45]:
```
Xnew = pandas.DataFrame(
    {
        "Diameter": [15],
        "Ratio": [65],
        "Terrain": [0],
        "Weight": [25],
        "Tread": [225],
        "WW": [0],
        "OWL": [0],
        "H": [0],
        "V": [0],
        "Z": [0],
        "Volume": [300],
```

```
    }
)

TireReg.prediction_intervals(Xnew, alpha=0.25)
```

[45]:

	Predicted	Lower	Upper
0	88.845747	80.661586	97.029907

We obtain a predicted price of \$88.85 and a 75% prediction interval of [\$80.66,\$97.02].

When you are unsure about keeping a variable, it can be useful to sensitivity analysis. Let's estimate the model without the Tread variable and redo prediction. We see the prediction and prediction interval change little. This is reassuring for the design team.

[46]:

```
Xnew = pandas.DataFrame(
    {
        "Diameter": [15],
        "Ratio": [65],
        "Terrain": [0],
        "Weight": [25],
        "WW": [0],
        "OWL": [0],
        "H": [0],
        "V": [0],
        "Z": [0],
        "Volume": [300],
    }
)

TireReg = tprstats.model(
    "cs", "Price~Diameter+Ratio+Terrain+Weight+Volume+WW+OWL+H+V+Z",␣
 ↪Tire_Pricing
)
TireReg.prediction_intervals(Xnew, alpha=0.25)
```

[46]:

	Predicted	Lower	Upper
0	88.025207	79.862866	96.187548

In the next chapter, we will study a test of functional form called the Ramsey test. We will apply the test to the tire pricing model of our alum, and we will find that the functional form fits the data well.

1.9 Summary

In this this chapter, we introduced the multiple regression model. We worked though applications to familiarize you with estimation and interpretation. We also introduced indicator variables, and we saw that such variables can be very quite valuable in building a regression model. As an application, we studied a model developed by an alum to aid his company in negotiating prices with tire suppliers. This is an exceptionally well crefted regression model.

The chapter also introduced some diagnostics to evaluate a multiple regression model. I suggest that you review the learning objectives presented at the beginning of this chapter to guide your review of material presented in this chapter.